

Symmetric Matrices

Hessian matrices.

为什么在优化问题中经常出现对称矩阵?

定义: Coindex.

A symmetric, The index of A $\text{Ind}(A)$, is defined to be the number of negative eigenvalues (multiplicity counted).

The coindex of A , $\text{Coind}(A)$, is the number of positive eigenvalues.

Hessian 矩阵: 目标函数 $f(x)$ 的二阶偏导数矩阵 $\nabla^2 f(x)$, 用于判断极值类型和收敛性.